Exercise J - A

1. The value of $\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right)$ is : **[JEE(Adv.)2013]**

$\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right)$ का मान है : **[JEE(Adv.)2013]**

(A) $\frac{23}{25}$ (B) $\frac{25}{23}$ (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

Ans. B

Sol. $\cot\sum_{n=1}^{23}\cot^{-1}(1+2+4+6+......+2n)$ $\Rightarrow$ $\cot\Sigma\cot^{-1}(1+n(n+1))$

$\cot\Sigma\tan^{-1}\frac{(n+1)-n}{1+n(n+1)}$ $\Rightarrow$ $\cot\sum_{n=1}^{23}\left(\tan^{-1}(n+1)-\tan^{-1}n\right)$

$\cot(\tan^{-1}24-\tan^{-1}1)$ $\Rightarrow$ $\cot\left(\tan^{-1}\frac{24-1}{1+24}\right)$ $\Rightarrow$ $\cot\left(\cot^{-1}\frac{25}{23}\right)=\frac{25}{23}$

2. Match Column I with Column II and select the correct answer using the code given below the columns : **[JEE(Adv.)2013]**

Column- I	**Column - II**
p. $\left(\frac{1}{y^2}\left(\frac{\cos\left(\tan^{-1}y\right)+y\sin\left(\tan^{-1}y\right)}{\cot\left(\sin^{-1}y\right)+\tan\left(\sin^{-1}y\right)}\right)^2+y^4\right)^{1/2}$ takes value	1. $\frac{1}{2}\sqrt{\frac{5}{3}}$
Q. If $\cos x+\cos y+\cos z=0=\sin x+\sin y+\sin z$ then possible value of $\cos\frac{x-y}{2}$ is	2. $\sqrt{2}$
R. If $\cos\left(\frac{\pi}{4}-x\right)\cos 2x+\sin x\sin 2x\sec x=\cos x\sin 2x\sec x$ $+\cos\left(\frac{\pi}{4}+x\right)\cos 2x$ then possible value of sec x is	3. $\frac{1}{2}$
S. If $\cot(\sin^{-1}\sqrt{1-x^2})=\sin(\tan^{-1}(x\sqrt{6})), x\neq 0$ then possible value of x is	4. 1

कॉलम I को कॉलम II से सुमेलित कीजिए तथा सूचियों के नीचे दिए गए कोड का प्रयोग करके सही उत्तर चुनिये :

[JEE(Adv.)2013]

Column- I	**Column - II**
p. $\left(\frac{1}{y^2}\left(\frac{\cos\left(\tan^{-1}y\right)+y\sin\left(\tan^{-1}y\right)}{\cot\left(\sin^{-1}y\right)+\tan\left(\sin^{-1}y\right)}\right)^2+y^4\right)^{1/2}$ का मान है	1. $\frac{1}{2}\sqrt{\frac{5}{3}}$
Q. यदि $\cos x+\cos y+\cos z=0=\sin x+\sin y+\sin z$ तब $\cos\frac{x-y}{2}$ का सम्भावित मान है	2. $\sqrt{2}$
R. यदि $\cos\left(\frac{\pi}{4}-x\right)\cos 2x + \sin x\sin 2x\sec x=\cos x\sin 2x\sec x$ $+\cos\left(\frac{\pi}{4}+x\right)\cos 2x$ तब $\sec x$ का सम्भावित मान है	3. $\frac{1}{2}$
S. यदि $\cot(\sin^{-1}\sqrt{1-x^2})=\sin(\tan^{-1}(x\sqrt{6})), x\neq 0$ तब x का सम्भावित मान है	4. 1

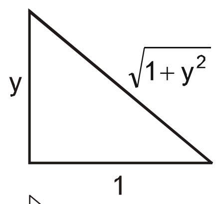
(A) P - 4; Q - 3; R - 1; S - 2 (B) P - 4; Q - 3; R - 2; S - 1

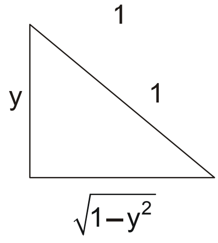
(C) P - 3; Q - 4; R - 2; S - 1 (D) P - 3; Q - 4; R - 1; S - 2

Ans. B

Sol. (P) $$\left(\frac{1}{y^2}\left(\frac{\cos(\tan^{-1}y)+y\sin(\tan^{-1}y)}{\cot(\sin^{-1}y)+\tan(\sin^{-1}y)}\right)^2+y^4\right)^{1/2}$$

$$=\left[\frac{1}{y^2}\left[\frac{\left(\frac{1}{\sqrt{1+y^2}}+\frac{y.y}{\sqrt{1+y^2}}\right)}{\left(\frac{\sqrt{1-y^2}}{y}+\frac{y}{\sqrt{1-y^2}}\right)}\right]^2+y^4\right]^{1/2}$$





$$=\left(\frac{1}{y^2}.y^2(1-y^4)+y^4\right)^{1/2}=1$$

Ans. 4

(Q) $\cos x+\cos y=-\cos z$

$\sin x+\sin y=-\sin z$ square and add वर्ग करके जोडने पर

$2+2\cos(x-y)=1$

$\Rightarrow \cos(x-y)=-1/2$

$\Rightarrow 2\cos^2\left(\frac{x-y}{2}\right)-1=-1/2, \Rightarrow \cos\left(\frac{x-y}{2}\right)=1/2$

Ans. 3

(R) $\cos 2x\left(\cos\left(\frac{\pi}{4}-x\right)-\cos\left(\frac{\pi}{4}+x\right)\right)+2\sin^2 x = 2\sin x\cos x$

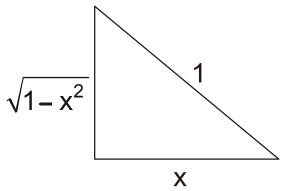
$\cos 2x\,(\sqrt{2}\sin x)+2\sin^2 x = 2\sin x\cos x \Rightarrow \sqrt{2}\sin x\,[\cos 2x+\sqrt{2}\sin x-\sqrt{2}\cos x]=0$

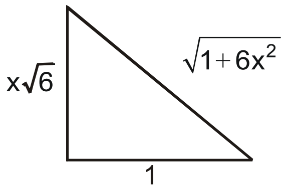
या तो Either $\sin x = 0$ OR या $\cos^2 x-\sin^2 x = \sqrt{2}(\cos x-\sin x)$

$\boxed{\sec x = 1}$ OR या $\cos x = \sin x \Rightarrow \boxed{\sec x = \sqrt{2}}$

Ans. 2 OR या 4

(S) $\cot(\sin^{-1}\sqrt{1-x^2}) = \sin(\tan^{-1}(x\sqrt{6}))$





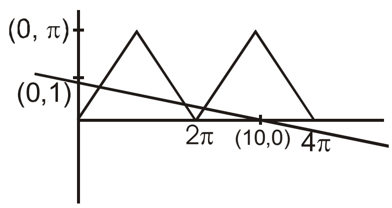
$$\frac{x}{\sqrt{1-x^2}} = \frac{x\sqrt{6}}{\sqrt{1+6x^2}} \Rightarrow 1+6x^2 = 6-6x^2 \Rightarrow 12x^2 = 5$$

$$x = \sqrt{\frac{5}{12}} = = \frac{1}{2}\sqrt{\frac{5}{3}}$$

3. Let $f : [0, 4\pi] \to [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is : **[JEE(Adv.)2014]**

माना फलन $f : [0, 4\pi] \to [0, \pi]$ इस प्रकार की $f(x) = \cos^{-1}(\cos x)$ है। अन्तराल $x \in [0, 4\pi]$ में स्थित बिन्दुओं की संख्या जिसके लिए समीकरण $f(x) = \frac{10-x}{10}$ सन्तुष्ट होगा : **[JEE(Adv.)2014]**

Ans. 3



Sol. $f(x) = \cos^{-1}(\cos x) \in [0, 4\pi]$ & $f(x) = \frac{10-x}{10} = 1-\frac{x}{10}$

so, 3 solution. इसलिए 3 हल

4. If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$ where the inverse trigonometric functions take only the principal values, then the correct option (s) is (are) : **[JEE(Adv.)2015]**

यदि $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ और $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, जहाँ प्रतिलोम त्रिकोणमितीय फलन (inverse trigonometric functions) केवल मुख्य मान (principal values) ही लेते हैं, तब सही कथन है (हैं) : **[JEE(Adv.)2015]**

(A) $\cos\beta > 0$ (B) $\sin\beta < 0$ (C) $\cos(\alpha + \beta) > 0$ (D) $\cos\alpha < 0$

Ans. BCD

Sol. $\alpha = 3\sin^{-1}\frac{6}{11} > 3\sin^{-1}\frac{6}{12}$ and $\beta = 3\cos^{-1}\frac{4}{9} > 3\cos^{-1}\frac{4}{8}$

$\Rightarrow \quad \alpha > \frac{\pi}{2}$ & $\quad \beta > \pi$

$\Rightarrow \quad \alpha + \beta > \frac{3\pi}{2}$

5. The number of real solutions of the equation $\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x\sum_{i=1}^{\infty}\left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty}\left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty}(-x)^i\right)$ lying in the interval $\left(-\frac{1}{2},\frac{1}{2}\right)$ is _____. (Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume values in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.) **[JEE(Adv.)2018]**

समीकरण $\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x\sum_{i=1}^{\infty}\left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty}\left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty}(-x)^i\right)$ के उन वास्तविक हलों की संख्या जो अन्तराल $\left(-\frac{1}{2},\frac{1}{2}\right)$ में विद्यमान हैं, है _____. (यहाँ प्रतिलोम त्रिकोणमितीय फलन $\sin^{-1}x$ और $\cos^{-1}x$ क्रमश: $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ व $[0, \pi]$ में मान धारण करते हैं।) **[JEE(Adv.)2018]**

Ans. 2

Sol. $$\text{Lim}_{n\to\infty}\sin^{-1}\left(\sum_{i=1}^{n} x^{i+1} - x\sum_{i=1}^{n}\left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \text{Lim}_{n\to\infty}\cos^{-1}\left(\sum_{i=1}^{n}\left(-\frac{x}{2}\right)^i - \sum_{i=1}^{n}(-x)^i\right)$$

$$\left(\frac{x^2}{1-x} - x\frac{\frac{x}{2}}{1-\frac{x}{2}}\right) = \frac{x}{1+x} + \frac{\left(-\frac{x}{2}\right)}{1+\frac{x}{2}}$$

$$\frac{x^2}{1-x}-\frac{x^2}{2-x}=\frac{x}{1+x}-\frac{x}{2+x}$$

$$\frac{x^2}{1-x}-\frac{x}{1+x}=\frac{x^2}{2-x}-\frac{x}{2+x}$$

$$\frac{x(1+x)-(1-x)}{1-x^2}=\frac{2x+x^2-2+x}{4-x^2} \quad \text{or } x=0$$

$$\frac{x^2+2x-1}{1-x^2}=\frac{x^2+3x-2}{4-x^2}$$

$\Rightarrow \quad x^3+2x^2+5x-2=0$

Let $\quad f(x)=x^3+2x^2+5x-2$

$f'(x)>0$

$f(0)=-2$ and $f(1/2)=9/8$ so one root in $\left(0,\frac{1}{2}\right)$

$\Rightarrow \quad$ 2 roots

6. For any positive integer n, define $f_n:(0,\infty)\to R$ as $f_n(x)=\sum_{j=1}^{n}\tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right)$ for all $x\in(0,\infty)$.

(Here, the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$). Then, which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) $\sum_{j=1}^{5}\tan^2\left(f_j(0)\right)=55$

(B) $\sum_{j=1}^{10}\left(1+f_j'(0)\right)\sec^2\left(f_j(0)\right)=10$

(C) For any fixed positive integer n, $\lim_{x\to\infty}\tan\left(f_n(x)\right)=\frac{1}{n}$

(D) For any fixed positive integer n, $\lim_{x\to\infty}\sec^2\left(f_n(x)\right)=1$

किसी भी धनात्मक पूर्णांक n के लिये, $f_n:(0,\infty)\to R$, $f_n(x)=\sum_{j=1}^{n}\tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right)$ सभी $x\in(0,\infty)$ के लिये, ($\tan^{-1}x$, $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ में मान धारण करता है।) तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) $\sum_{j=1}^{5}\tan^2\left(f_j(0)\right)=55$

(B) $\sum_{j=1}^{10}\left(1+f_j'(0)\right)\sec^2\left(f_j(0)\right)=10$

(C) किसी भी नियत धनात्मक पूर्णांक n के लिये, $\lim\limits_{x\to\infty}\tan\left(f_n(x)\right)=\frac{1}{n}$

(D) किसी भी नियत धनात्मक पूर्णांक n के लिये, $\lim\limits_{x\to\infty}\ \sec^2\left(f_n(x)\right)=1$

Ans. D

Sol. $f_n(x)=\sum\limits_{j=1}^{n}\tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right)$

$f_n(x)=\tan^{-1}(x+n)-\tan^{-1}(x)$ $\quad\Rightarrow\quad f_n'(x)=\frac{1}{1+(x+n)^2}-\frac{1}{1+x^2}$

$f_n(0)=\tan^{-1}(n)$ $\quad\Rightarrow\quad \tan^2\left(\tan^{-1}n\right)=n^2$

(A) $\sum\limits_{j=1}^{5}\tan^2\left(f_j(0)\right)=\sum\limits_{j=1}^{5}j^2=\frac{5.6.11}{6}=55$

(B) $f_n'(0)=\frac{1}{1+n^2}-1 \quad\Rightarrow\quad 1+f_n(0)=\frac{1}{1+n^2}$

$\sec^2\left(f_n(0)\right)=\sec^2\left(\tan^{-1}(n)\right)=1+n^2$

Hence $\left(1+f_n'(0)\right)\cdot\sec^2\left(f_n(0)\right)=\left(\frac{1}{1+n^2}\right)\left(1+n^2\right)=1$

So $\sum\limits_{i=1}^{10}\left(1+f_{ii}(0)\right)\sec^2\left(f_i(0)\right)=\sum\limits_{i=1}^{10}1=10$

$\lim\limits_{x\to\infty}f_n(x)=\lim\limits_{x\to\infty}\tan^{-1}\left(\frac{n}{1+x(n+x)}\right)=0$

$\lim\limits_{x\to\infty}\tan\left(f_n(x)\right)=0$ and $\lim\limits_{x\to\infty}\sec^2\left(f_n(x)\right)=1$

7. Let $E_1=\left\{x\in\mathbb{R}:x\neq1 \text{ and } \frac{x}{x-1}>0\right\}$ and $E_2=\left\{x\in E_1:\sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)\text{ is a real number}\right\}$.

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes values in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.) Let $f: E_1\to\mathbb{R}$ be the function defined by $f(x)=\log_e\left(\frac{x}{x-1}\right)$ and $g: E_2\to\mathbb{R}$ be the function defined by $g(x)=\sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$

Column-I		Column-II
P.	The range of f is	1. $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
Q.	The range of g contains	2. $(0, 1)$
R.	The domain of f contains	3. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
S.	The domain of g is	4. $(-\infty, 0) \cup (0, \infty)$
		5. $\left(-\infty, \frac{e}{e-1}\right]$
		6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{1-e}\right]$

The correct option is : **[JEE(Adv.)2018]**

माना कि $E_1 = \left\{x \in \mathbb{R} : x \neq 1 \text{ और } \frac{x}{x-1} > 0\right\}$ और $E_2 = \left\{x \in E_1 : \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right) \text{ एक वास्तविक संख्या है।}\right\}$.

(यहाँ प्रतिलोम त्रिकोणमितीय फलन (inverse trigonometric function) $\sin^{-1}x$, $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ में मान धारण करता है।) माना कि फलन $f : E \rightarrow \mathbb{R}$, $f(x) = \log_e\left(\frac{x}{x-1}\right)$ के द्वारा परिभाषित है और फलन $g : E_2 \rightarrow \mathbb{R}$, $g(x) = \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$ के द्वारा परिभाषित है।

Column-I		Column-II
P.	f का परिसर है	1. $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
Q.	g के परिसर में समाहित है	2. $(0, 1)$
R.	f के प्रान्त में समाहित है	3. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
S.	g का प्रानत है	4. $(-\infty, 0) \cup (0, \infty)$
		5. $\left(-\infty, \frac{e}{e-1}\right]$
		6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{1-e}\right]$

दिए हुए विकल्पों में से सही विकल्प है : **[JEE(Adv.)2018]**

(A) P - 4; Q - 2; R - 1; S - 1
(B) P - 3; Q - 3; R - 6; S - 5
(C) P - 4; Q - 2; R - 1; S - 6
(D) P - 4; Q - 3; R - 6; S - 5

Ans. A

Sol. $E_1 \dfrac{x}{x-1} > 0 \quad \Rightarrow \quad x \in (-\infty, 0) \cup (1, \infty)$

$E_2 : -1 \le \ell n\left(\dfrac{x}{x-1}\right) \le 1 \quad \Rightarrow \quad \dfrac{1}{e} \le \dfrac{x}{x-1} \le e \quad \Rightarrow \quad \dfrac{1}{e} \le 1 + \dfrac{1}{x-1} \le e$

$\dfrac{1}{e} - 1 \le \dfrac{1}{x-1} \le e - 1 \quad \Rightarrow \quad (x-1) \in \left(-\infty, \dfrac{e}{1-e}\right] \cup \left[\dfrac{1}{1-e}, \infty\right)$

$x \in \left(-\infty, \dfrac{1}{e-1}\right] \cup \left[\dfrac{e}{e-1}, \infty\right)$

Now $\dfrac{x}{x-1} \in (0, \infty) - \{1\} \; \forall \in E_1 \quad \Rightarrow \quad \ell n\left(\dfrac{x}{x-1}\right) \in (-\infty, \infty) - \{0\}$

$\sin^{-1}\left(\ell n\left(\dfrac{x}{x-1}\right)\right) \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right] - \{0\}$

8. For non negative integers n let $f(n) = \dfrac{\sum_{k=0}^{n} \sin\left(\frac{k+1}{n+2}\pi\right) \sin\left(\frac{k+2}{n+2}\pi\right)}{\sum_{k=0}^{n} \sin^2\left(\frac{k+1}{n+2}\pi\right)}$ Assuming $\cos^{-1} x$ takes values in $[0, \pi]$ which of the following options is/are correct : **[JEE(Adv.)2019]**

(A) $f(4) = \dfrac{\sqrt{3}}{2}$

(B) If $\alpha = \tan(\cos^{-1} f(6))$ then $\alpha^2 + 2\alpha - 1 = 0$

(C) $\lim\limits_{n \to \infty} f(n) = \dfrac{1}{2}$

(D) $\sin(7\cos^{-1} f(5)) = 0$

अऋणात्मक पूर्णांकों (non negative integers) n के लिए माना कि $f(n) = \dfrac{\sum_{k=0}^{n} \sin\left(\frac{k+1}{n+2}\pi\right) \sin\left(\frac{k+2}{n+2}\pi\right)}{\sum_{k=0}^{n} \sin^2\left(\frac{k+1}{n+2}\pi\right)}$ माना कि $\cos^{-1} x$ का मान $[0, \pi]$ में है, तब निम्न में से कौन सा (से) विकल्प सही है (हैं) : **[JEE(Adv.)2019]**

(A) $f(4) = \dfrac{\sqrt{3}}{2}$

(B) यदि $\alpha = \tan(\cos^{-1} f(6))$ then $\alpha^2 + 2\alpha - 1 = 0$

(C) $\lim\limits_{n \to \infty} f(n) = \dfrac{1}{2}$

(D) $\sin(7\cos^{-1} f(5)) = 0$

Ans. ABD

Sol. $$f(n)=\frac{\sum_{k=0}^{n}2\sin\left((k+1)\frac{\pi}{n+2}\right)\sin\left((k+2)\frac{\pi}{n+2}\right)}{\sum_{k=0}^{n}2\sin^2\left((k+1)\frac{\pi}{n+2}\right)}$$

$$\frac{\sum_{k=0}^{n}\left(\cos\frac{\pi}{n+2}-\cos\left((2k+3)\frac{\pi}{n+2}\right)\right)}{\sum_{k=0}^{n}1-\cos\left((2k+2)\frac{\pi}{n+2}\right)}$$

$$f(n)=\frac{(n+1)\cos\left(\frac{\pi}{n+2}\right)-\sum_{k=0}^{n}\cos\left((2k+3)\frac{\pi}{n+2}\right)}{(n+1)-\sum_{k=0}^{n}\cos\left((2k+2)\frac{\pi}{n+2}\right)}\quad\text{..................(1)}$$

$$\sum_{k=0}^{n}\cos\left((2k+3)\frac{\pi}{n+2}\right)=\cos\left(\frac{3\pi}{n+2}\right)+\cos\frac{5\pi}{n+2}+\cos\frac{7\pi}{n+2}+.....+\cos(2n+3)\frac{\pi}{n+2}$$

$$=\frac{\sin\left(\frac{(n+1)\frac{2\pi}{n+2}}{2}\right)}{\sin\left(\frac{\pi}{n+2}\right)}.\cos\left(\frac{\frac{3\pi}{n+2}+(2n+3)\frac{\pi}{n+2}}{2}\right)$$

$$=\frac{\sin\left((n+1)\frac{\pi}{n+2}\right)}{\sin\left(\frac{\pi}{n+2}\right)}.\cos\left((n+3)\frac{\pi}{n+2}\right)\ [\sin(\pi-\theta)=\sin\theta]$$

$$=\cos\left((n+3)\frac{\pi}{n+2}\right)=\cos\left(\pi+\frac{\pi}{n+2}\right)=-\cos\left(\frac{\pi}{n+2}\right)$$

$$\sum_{k=0}^{h}\cos\left((2k+2)\frac{\pi}{n+2}\right)=\cos\left(\frac{2\pi}{n+2}\right)+\cos\frac{4\pi}{n+2}+.....+\cos\left((2n+2)\frac{\pi}{n+2}\right)$$

$$=\frac{\sin\left((n+1)\frac{\pi}{n+2}\right)}{\sin\left(\frac{\pi}{n+2}\right)}\cos\left((n+2)\frac{\pi}{n+2}\right)$$

$= \cos\pi = -1$

$$f(n) = \frac{(n+1)\cos\left(\frac{\pi}{n+2}\right) - \left(-\cos\frac{\pi}{n+2}\right)}{n+1-(-1)}$$

$$f(n) = \cos\frac{(n+2)\cos\left(\frac{\pi}{n+2}\right)}{n+2}$$

$$f(n) = \cos\left(\frac{\pi}{n+2}\right)$$

Option: (1) $f(4) = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

(2) $f(6) = \cos^{-1}(f(6) = \frac{\pi}{8}$

$\alpha = \tan\frac{\pi}{8}$

$\alpha^2 + 2\alpha - 1 = 0$

$2\alpha = 1 - \alpha^2$

$\frac{2\alpha}{1-\alpha^2} = 1$

$\tan\left(\frac{\pi}{8}\times 2\right) = 1$

$\tan 45° = 1$ true

(3) $n \to \infty$

$f(n) \to 1$

(4) $f(5) = \cos\left(\frac{\pi}{7}\right)$

$\cos^{-1}(f(5)) = \frac{\pi}{7}$

$\sin\left(7\times\frac{\pi}{7}\right) = \sin\pi = 0$

9. The value of $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4},\frac{3\pi}{4}\right]$ equals :

[JEE(Adv.)2019]

अन्तराल $\left[-\frac{\pi}{4},\frac{3\pi}{4}\right]$ में $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$ का मान बराबर है :

[JEE(Adv.)2019]

Ans. 0

Sol. $$\frac{1}{4}\sum_{k=0}^{10}\frac{1}{\cos\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\cos\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)}$$

$$=\frac{1}{4}\sum_{k=0}^{10}\frac{\sin\left(\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)-\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\right)}{\cos\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\cos\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)}$$

$$=\frac{1}{4}\sum_{k=0}^{10}\left(\tan\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)-\tan\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\right)$$

$$=\frac{1}{4}\left(\tan\left(\frac{11\pi}{2}+\frac{7\pi}{12}\right)-\tan\frac{7\pi}{12}\right)$$

$$=\frac{1}{4}\left(\tan\left(\frac{\pi}{12}\right)-\tan\frac{7\pi}{12}\right)$$

$$=\frac{1}{4}\left(\tan\frac{\pi}{12}+\cot\frac{\pi}{12}\right)$$

$$=\frac{1}{4}2\operatorname{cosec}\left(2\times\frac{\pi}{12}\right)$$

$$=\frac{1}{4}\times 2\operatorname{cosec}\left(\frac{\pi}{6}\right)$$

$$=\frac{1}{4}\times 2\times 2=1$$

$\sec^{-1}(1)=0$

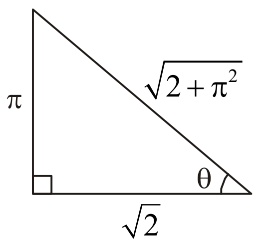
10. Considering only the principal values of the inverse trigonometric functions, the value of

$\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi}$ is ____. **[JEE(Adv.)2022]**

प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मानों पर विचार करते हुए, $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi}$ का मान ज्ञात कीजिए। **[JEE(Adv.)2022]**

Ans. $\frac{3\pi}{4}$

Sol. $\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}=\tan^{-1}\frac{\pi}{\sqrt{2}}$



$$\sin^{-1}\left(\frac{2\sqrt{2}\pi}{2+\pi^2}\right)=\sin^{-1}\left(\frac{2\times\frac{\pi}{\sqrt{2}}}{1+\left(\frac{\pi}{\sqrt{2}}\right)^2}\right)$$

$$=\pi-2\tan^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$$

$$\left(\text{As, }\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\pi-2\tan^{-1}x, x\geq 1\right)$$

and $\tan^{-1}\frac{\sqrt{2}}{\pi}=\cot^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$

$$\therefore \text{Expression}=\frac{3}{2}\left(\tan^{-1}\frac{\pi}{\sqrt{2}}\right)+\frac{1}{4}\left(\pi-2\tan^{-1}\frac{\pi}{\sqrt{2}}\right)+\cot^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$$

$$=\left(\frac{3}{2}-\frac{2}{4}\right)\tan^{-1}\frac{\pi}{\sqrt{2}}+\frac{\pi}{4}+\cot^{-1}\frac{\pi}{\sqrt{2}}$$

$$=\left(\tan^{-1}\frac{\pi}{\sqrt{2}}+\cot^{-1}\frac{\pi}{\sqrt{2}}\right)+\frac{\pi}{4}$$

$$=\frac{\pi}{2}+\frac{\pi}{4}=\frac{3\pi}{4}$$

$$=2.35 \text{ or } 2.36$$

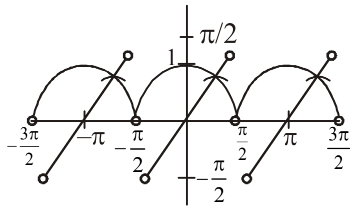
11. Let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, for $x \in R$. Then the number of real solutions of the equation $\sqrt{1+\cos(2x)} = \sqrt{2}\tan^{-1}(\tan x)$ in the set $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ is equal to________.

[JEE(Adv.)2023]

माना कि $x \in R$ के लिए $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ है। तब समुच्चय $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ में समीकरण $\sqrt{1+\cos(2x)} = \sqrt{2}\tan^{-1}(\tan x)$ के वास्तविक हलों की संख्या है ________ **[JEE(Adv.)2023]**

Ans. 3

Sol. $\sqrt{1+\cos 2x} = \sqrt{2}\tan^{-1}(\tan x)$



$\sqrt{2\cos^2 x} = \sqrt{2}\tan^{-1}(\tan x)$

$\sqrt{2}|\cos x| = \sqrt{2}\tan^{-1}(\tan x)$

$y = |\cos x|$ $\qquad\qquad$ $y = \tan^{-1}(\tan x)$

3 Solution

12. For any $y \in R$, let $\cot^{-1}(y) \in (0, \pi)$ and $\tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the sum of all the solutions of the equation

$\tan^{-1}\left(\frac{6y}{9-y^2}\right) + \cot^{-1}\left(\frac{9-y^2}{6y}\right) = \frac{2\pi}{3}$ for $0 < |y| < 3$, is equal to : **[JEE(Adv.)2023]**

किसी $y \in R$, के लिए माना कि $\cot^{-1}(y) \in (0, \pi)$ एवं $\tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ है। तब समीकरण

$\tan^{-1}\left(\frac{6y}{9-y^2}\right) + \cot^{-1}\left(\frac{9-y^2}{6y}\right) = \frac{2\pi}{3}$, जहाँ $0 < |y| < 3$ है, के सभी हलों का योगफल है : **[JEE(Adv.)2023]**

(A) $2\sqrt{3} - 3$ (B) $3 - 2\sqrt{3}$ (C) $4\sqrt{3} - 6$ (D) $6 - 4\sqrt{3}$

Ans. C

Sol. Case-I : $y \in (-3, 0)$

$$\tan^{-1}\left(\frac{6y}{9-y^2}\right) + \pi + \tan^{-1}\left(\frac{6y}{9-y^2}\right) = \frac{2\pi}{3}$$

$$2\tan^{-1}\left(\frac{6y}{9-y^2}\right) = -\frac{\pi}{3}$$

$y^2 - 6\sqrt{3}y - 9 = 0 \Rightarrow y = 3\sqrt{3} - 6 (\because y \in (-3, 0))$

Case-I : $y \in (0,3)$

$$2\tan^{-1}\left(\frac{6y}{9-y^2}\right) = \frac{2\pi}{3} \Rightarrow \sqrt{3}y^2 + 6y - 9\sqrt{3} = 0$$

$y = \sqrt{3}$ or $y = -3\sqrt{3}$ (rejected)

$\text{sum} = \sqrt{3} + 3\sqrt{3} - 6 = 4\sqrt{3} - 6$

13. Considering only the principal values of the inverse trigonometric functions, the value of $\tan\left(\sin^{-1}\left(\frac{3}{5}\right) - 2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$ is : **[JEE(Adv.)2024]**

प्रतिलोम त्रिकोणमितीय फलनों (inverse trigonometric functions) के केवल मुख्य मानों (principal values) को ध्यान में रखते हुए, $\tan\left(\sin^{-1}\left(\frac{3}{5}\right) - 2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$ का मान है : **[JEE(Adv.)2024]**

(A) $\frac{7}{24}$ (B) $\frac{-7}{24}$ (C) $\frac{-5}{24}$ (D) $\frac{5}{24}$

Ans. B

Sol. $\tan\left(\sin^{-1}\left(\frac{3}{5}\right) - 2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$

$\sin^{-1}\left(\frac{3}{5}\right) = \theta$ $\qquad$ $2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \phi$

$\sin\theta = \frac{3}{5}$ $\qquad$ $\cos\frac{\phi}{2} = \frac{2}{\sqrt{5}}$

$\tan\theta = \frac{3}{4}$ $\qquad$ $\tan\frac{\phi}{2} = \frac{1}{2}$

$$\tan\phi = \frac{2\tan\frac{\phi}{2}}{1-\tan^2\frac{\phi}{2}}$$

$$\tan\phi = \frac{2\left(\frac{1}{2}\right)}{1-\frac{1}{4}}$$

$$\tan\phi = \frac{4}{3}$$

$\tan(\phi - \theta)$

$$\frac{\tan\theta - \tan\phi}{1+\tan\theta\tan\phi}$$

$$\frac{\frac{3}{4}-\frac{4}{3}}{1+1} = \frac{9-16}{24}$$

$$= -\frac{7}{24}$$

14. The total number of real solutions of the equation $\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}\sin^{-1}\left(\frac{6\tan\theta}{9+\tan^2\theta}\right)$ is

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\tan^{-1}x$ assume values in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, respectively)

[JEE(Adv.)2025]

समीकरण (equation) $\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}\sin^{-1}\left(\frac{6\tan\theta}{9+\tan^2\theta}\right)$ के कुल वास्तविक हलों की संख्या (total number of real solutions) है

(यहाँ प्रतिलोम त्रिकोणमितीय फलनों, (inverse trigonometric functions) $\sin^{-1}x$ तथा $\tan^{-1}x$ के मान क्रमशः $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ और $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ में हैं।)

[JEE(Adv.)2025]

(A) 1 (B) 2 (C) 3 (D) 5

Ans. C

Sol. $\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}\sin^{-1}\left(\frac{2\left(\frac{\tan\theta}{3}\right)}{1+\left(\frac{\tan\theta}{3}\right)^2}\right)$

Let $\frac{\tan\theta}{3} = \tan\alpha$

$\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}\sin^{-1}(\sin 2\alpha)$

Case-I : $2\alpha \in \left(-\pi, -\frac{\pi}{2}\right)$

$\alpha \in \left(-\frac{\pi}{2}, -\frac{\pi}{4}\right)$

$\tan\alpha \in (-\infty, -1)$

$\frac{\tan\theta}{3} \in (-\infty, -1)$

$\tan\theta \in (-\infty, -3)$

$\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}(-\pi - 2\alpha)$

$\theta - \frac{\pi}{2} = \tan^{-1}(2\tan\theta) + \alpha$

$\tan\left(\theta - \frac{\pi}{2}\right) = \frac{2\tan\theta + \tan\alpha}{1 - 2\tan\theta \cdot \tan\alpha}$

$-\frac{1}{\tan\theta} = \frac{2\tan\theta + \frac{\tan\theta}{3}}{1 - \frac{2\tan^2\theta}{3}}$

$\frac{-1}{\tan\theta} = \frac{7\tan\theta}{3 - 2\tan^2\theta}$

$5\tan^2\theta = -3$

No solution.

Case-II : $2\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\alpha \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

$\tan\theta \in [-3, 3]$

$\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}(2\alpha)$

$\tan\theta = \frac{2\tan\theta - \tan\alpha}{1 + 2\tan\theta\tan\alpha}$

$\tan\theta = \frac{5\tan\theta}{3 + 2\tan^2\theta}$ $\qquad \left(\tan\alpha = \frac{\tan\theta}{3}\right)$

$\tan\theta = 0$ or $\tan^2\theta = 1$

$\tan\theta = \pm 1$

3 solutions

Case-III : $2\alpha \in \left(\frac{\pi}{2}, \pi\right)$

$\alpha \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$\tan\theta \in (3, \infty)$

$\theta = \tan^{-1}(2\tan\theta) - \frac{1}{2}(\pi - 2\alpha)$

$\frac{\pi}{2} + \theta = \tan^{-1}(2\tan\theta) + \alpha$

$\tan\left(\frac{\pi}{2} + \theta\right) = \frac{2\tan\theta + \tan\alpha}{1 - 2\tan\theta\tan\alpha}$

$\frac{-1}{\tan\theta} = \frac{7\tan\theta}{3 - 2\tan^2\theta}$ $\qquad \left(\tan\alpha = \frac{\tan\theta}{3}\right)$

$5\tan^2\theta = -3$

No solution

Total 3 solutions.

Exercise C - I

1. The principal value of $\sin^{-1}\left[3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(5\pi+\alpha)\right]\right]$ is equal to :

$\sin^{-1}\left[3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(5\pi+\alpha)\right]\right]$ का मुख्य मान होगा :

(A) $-\frac{2\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{3}$ (D) $\frac{\pi}{2}$

Ans. D

Sol. $\sin^{-1}\left[3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(5\pi+\alpha)\right]\right]$

$\Rightarrow \sin^{-1}\left[3\cos^4\alpha+3\sin^4\alpha-2\cos^6\alpha-2\sin^6\alpha\right]$

$\Rightarrow \sin^{-1}\left[3\left(1-2s^2c^2\right)-2\left(1-3s^2c^2\right)\right]$

$\Rightarrow \sin^{-1}\left[3-6s^2c^2-2+6s^2c^2\right]$

$=\sin^{-1}(1)=\frac{\pi}{2}$

2. If $\sin\left(\cot^{-1}(x+1)\right)=\cos(\tan^{-1}x)$, then $x=$

यदि $\sin\left(\cot^{-1}(x+1)\right)=\cos(\tan^{-1}x)$, तब $x=$

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 0 (D) $\frac{9}{4}$

Ans. A

Sol. $\sin\left(\cot^{-1}(x+1)\right)=\cos(\tan^{-1}\frac{x}{1})$

$$\sin\left(\sin^{-1}\left(\frac{1}{\sqrt{x^2+2x+2}}\right)\right)=\cos\left(\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)$$

$$\Rightarrow \frac{1}{\sqrt{x^2+2x+2}} = \frac{1}{\sqrt{1+x^2}}$$

$1 + x^2 = x^2 + 2x + 2$

$$x = \frac{-1}{2}$$

3. Sum of series $\cot^{-1}\left(\frac{5}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{9}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{15}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{23}{\sqrt{3}}\right)+.....$ upto infinite terms is equal to :

श्रेणी $\cot^{-1}\left(\frac{5}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{9}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{15}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{23}{\sqrt{3}}\right)+.....$ के अनन्त पदों का योग होगा :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Ans. B

Sol. $\tan^{-1}\left(\frac{\sqrt{3}}{5}\right)+\tan^{-1}\left(\frac{\sqrt{3}}{9}\right)+\tan^{-1}\left(\frac{\sqrt{3}}{15}\right)+\tan\left(\frac{\sqrt{3}}{13}\right)+........\infty$

$$T_r = \tan^{-1}\left(\frac{\sqrt{3}}{r^2+r+3}\right)$$

$$T_r = \tan^{-1}\left(\frac{\frac{1}{\sqrt{3}}}{1+\frac{r}{\sqrt{3}}\frac{(r+1)}{\sqrt{3}}}\right)$$

$$T_r = \tan^{-1}\left(\frac{\frac{r+1}{\sqrt{3}}-\frac{r}{3}}{1+\frac{r}{\sqrt{3}}\frac{(r+1)}{\sqrt{3}}}\right)$$

$$T_r = \tan^{-1}\left(\frac{r+1}{\sqrt{3}}\right)-\tan^{-1}\left(\frac{r}{\sqrt{3}}\right)$$

$$T_1 = \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$T_2 = \tan^{-1}\left(\frac{3}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

$$\vdots \qquad \vdots \qquad\qquad \vdots$$

$$T_n = \tan^{-1}\left(\frac{n+1}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{n}{\sqrt{3}}\right)$$

$$S_n = \tan^{-1}\left(\frac{n+1}{\sqrt{3}}\right) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$S_\infty = \tan^{-1}(\infty) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{\pi}{2} - \frac{\pi}{6}$$

$$S_\infty = \frac{\pi}{3}$$

4. If $2\tan^{-1}\left(\frac{1+x}{1-x}\right) = 5\tan^{-1}x$, then value of x is :

यदि $2\tan^{-1}\left(\frac{1+x}{1-x}\right) = 5\tan^{-1}x$, तो x का मान होगा :

(A) $\tan\left(\frac{1}{\sqrt{3}}\right)$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{\pi}{4}$ (D) $\sqrt{3}$

Ans. B

Sol. $2\tan^{-1}\left(\frac{1+x}{1-x}\right) = 5\tan^{-1}x$

$$2\tan^{-1}\left(\tan\left(\frac{\pi}{4} + \tan^{-1}x\right)\right) = 5\tan^{-1}x$$

$\tan^{-1}x = \theta$

$x = \tan\theta$

$$2\left(\frac{\pi}{4}+\theta\right)=5\theta$$

$$\frac{\pi}{2}+2\theta=5\theta$$

$$3\theta=\frac{\pi}{2}$$

$$\theta=\frac{\pi}{6}$$

$$x=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}$$

5. If $x \in (0,1)$, then solution of inequality $\sin(2\sin^{-1}x) > \sqrt{2}\,x$ is given by :

यदि $x \in (0,1)$ हो, तो असमिका $\sin(2\sin^{-1}x) > \sqrt{2}\,x$ का हल होगा :

(A) $x \in \left(\frac{1}{\sqrt{2}},\frac{\sqrt{3}}{2}\right)$ (B) $x \in \left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$ (C) $(0, 1)$ (D) $\left(0,\frac{1}{\sqrt{2}}\right)$

Ans. D

Sol. $\sin(2\sin^{-1}x) > \sqrt{2}\,x$

$x \in (0, 1)$ ....(1)

$\sin^{-1}x=\theta$

$\sin\theta=x$

$$\sin 2\theta > \sqrt{2}\sin\theta$$

$$\sin\theta\left(2\cos\theta-\sqrt{2}\right)>0$$

$$\underbrace{x}_{x>0}\left(\sqrt{1-x^2}-\frac{1}{\sqrt{2}}\right)>0$$

$$\sqrt{1-x^2}-\frac{1}{\sqrt{2}}>0$$

$$1-x^2>\frac{1}{2}$$

$$x^2<\frac{1}{2}$$

$|x| < \frac{1}{\sqrt{2}}$

$x \in \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ ....(2)

from (1) & (2)

$x \in \left(0, \frac{1}{\sqrt{2}}\right)$

6. If $\sin^{-1}(x^2) = \cos^{-1} x$, then $x^4 + x^2$ is :

यदि $\sin^{-1}(x^2) = \cos^{-1} x$ हो, तो $x^4 + x^2$ होगा :

(A) $\frac{1}{4}$ (B) 1 (C) $\frac{1}{2}$ (D) $\frac{2}{5}$

Ans. B

Sol. $\sin^{-1}(x^2) = \cos^{-1} x$

$\sin^{-1} x^2 = \sin^{-1}\sqrt{1-x^2}$

$x^2 = \sqrt{1-x^2}$

$x^4 = 1 - x^2$

$x^4 + x^2 = 1$

ONE OR MORE THAN ONE CORRECT TYPE

7. Which of the following functions represent identical graphs in x-y plane $\forall\, x \in (2, 3)$?

निम्न में से कौनसे फलन समरूप है $\forall\, x \in (2, 3)$?

(A) $y = \cos^{-1}\sqrt{3-x}$ (B) $y = \sin^{-1}\sqrt{x-2}$

(C) $y = \cot^{-1}\left(\sqrt{\frac{3-x}{x-2}}\right)$ (D) $y = \frac{1}{2}\sin^{-1}\left(2\sqrt{(3-x)(x-2)}\right)$

Ans. ABC

Sol. Let $\sin^{-1}\sqrt{x-2} = \alpha$ $\left(\text{where } \alpha \in \left(0, \frac{\pi}{2}\right)\right)$ ....(i)

then $(x-2) = \sin^2\alpha$ $\Rightarrow 3 - x = 1 - (x-2) = 1 - \sin^2\alpha = \cos^2\alpha$

$\therefore\ \alpha = \cos^{-1}\sqrt{3-x}$ ....(ii)

Also $\cot^2\alpha = \dfrac{\cos^2\alpha}{\sin^2\alpha} = \dfrac{3-x}{x-2}$ $\quad\therefore\ \alpha = \cot^{-1}\left(\sqrt{\dfrac{3-x}{x-2}}\right)$ ....(iii)

Also, $\sin 2\alpha = 2\sin\alpha\cos\alpha = 2\sqrt{x-2}\,\sqrt{3-x}$

$$\therefore\quad 2\alpha = \begin{cases} \sin^{-1}\left(2\sqrt{(x-2)(3-x)}\right), & \text{if } 0<\alpha<\dfrac{\pi}{4} \\ \pi-\sin^{-1}\left(2\sqrt{(x-2)(3-x)}\right), & \text{if } \dfrac{\pi}{4}\le\alpha<\dfrac{\pi}{2} \end{cases}$$

$$\Rightarrow \alpha = \begin{cases} \dfrac{1}{2}\sin^{-1}\left(2\sqrt{(x-2)(3-x)}\right), & \text{if } 0<\alpha<\dfrac{\pi}{4} \\ \dfrac{\pi}{2}-\dfrac{1}{2}\sin^{-1}\left(2\sqrt{(x-2)(3-x)}\right), & \text{if } \dfrac{\pi}{4}\le\alpha<\dfrac{\pi}{2} \end{cases} \text{......(iv)}$$

From (i), (ii), (iii) and (iv) it is clear that

$$y = \cos^{-1}\sqrt{3-x}\ ,\ y = \sin^{-1}\sqrt{x-2} \text{ and } y = \cot^{-1}\sqrt{\frac{3-x}{x-2}}$$

have identical graphs in the range $\left(0,\ \dfrac{\pi}{2}\right)$ but $y = \dfrac{1}{2}\sin^{-1}\left(2\sqrt{(3-x)(x-2)}\right)$ has range $\left(0,\ \dfrac{\pi}{4}\right)$

8. Let x_1, x_2, x_3, x_4 be four non zero numbers satisfying the equation $\tan^{-1}\dfrac{a}{x}+\tan^{-1}\dfrac{b}{x}+\tan^{-1}\dfrac{c}{x}+\tan^{-1}\dfrac{d}{x}=\dfrac{\pi}{2}$ then which of the following relation(s) hold good?

यदि x_1, x_2, x_3, x_4 चार अशून्य संख्याऐं समीकरण $\tan^{-1}\dfrac{a}{x}+\tan^{-1}\dfrac{b}{x}+\tan^{-1}\dfrac{c}{x}+\tan^{-1}\dfrac{d}{x}=\dfrac{\pi}{2}$ को सन्तुष्ट करते हैं, तब निम्न में से कौन कौन से सम्बन्ध सत्य होंगे?

(A) $\sum_{i=1}^{4} x_i = a+b+c+d$

(B) $\sum_{i=1}^{4} \dfrac{1}{x_i} = 0$

(C) $\prod_{i=1}^{4} x_i = abcd$

(D) $(x_1+x_2+x_3)(x_2+x_3+x_4)(x_3+x_4+x_1)(x_4+x_1+x_2) = abcd$

Ans. BCD

Sol. Let $\tan^{-1}\frac{a}{x} = \alpha \quad \Rightarrow \quad \tan\alpha = \frac{a}{x}$ etc.

$$\alpha + \beta + \gamma + \delta = \frac{\pi}{2}$$

$$\tan(\alpha + \beta + \gamma + \delta) = \tan\frac{\pi}{2}$$

$$\frac{S_1 - S_3}{1 - S_2 + S_4} = \infty \quad \Rightarrow \quad 1 - S_2 + S_4 = 0 \Rightarrow \quad S_4 - S_2 + 1 = 0$$

Now, $S_4 = \tan\alpha \cdot \tan\beta \cdot \tan\gamma \cdot \tan\delta = \frac{abcd}{x^4}$

$$S_2 = \sum \tan\alpha \cdot \tan\beta = \frac{\sum ab}{x^2}$$

$$\therefore \quad \frac{abcd}{x^4} - \frac{\sum ab}{x^2} + 1 = 0$$

$$x^4 - \sum ab\, x^2 + abcd = 0 \begin{array}{l} x_1 \\ x_2 \\ x_3 \\ x_4 \end{array}$$

$$\therefore \quad x_1 + x_2 + x_3 + x_4 = 0 \quad ....(1)$$

$$\sum x_1x_2x_3 = 0 \quad ....(2)$$

$$\sum x_1x_2x_3 = \underbrace{x_1x_2x_3x_4}_{\text{non zero}}\left[\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4}\right] = 0 \Rightarrow \quad (B)$$

$$x_1x_2x_3x_4 = abcd \Rightarrow \quad (C)$$

9. If $x^2 + 2x + n > 10 + \sin^{-1}(\sin 9) + \tan^{-1}(\tan 9)$ for all real x, then the possible value of n can be :

यदि $x^2 + 2x + n > 10 + \sin^{-1}(\sin 9) + \tan^{-1}(\tan 9)$ $\forall$ $x \in R$ तो n के सम्भव मान होंगे :

(A) 11 (B) 12 (C) 13 (D) 14

Ans. BCD

Sol. We have $x^2 + 2x + (n - 10) > 0 \;\; \forall\, x \in R$ [$\because$ $\sin^{-1}(\sin 9) = 3\pi - 9$ and $\tan^{-1}(\tan 9) = 9 - 3\pi$

$\therefore \quad \sin^{-1}(\sin 9) + \tan^{-1}(\tan 9) = 3\pi - 9 + 9 - 3\pi = 0$]

$\therefore$ Discriminant < 0

$\Rightarrow 4 - 4(n - 10) < 0 \Rightarrow 1 - n + 10 < 0 \Rightarrow n > 11$

10. Let $f : R \to R$ defined by $f(x) = \cos^{-1}(-\{-x\})$ where $\{x\}$ is fractional part function. Then which of the following is/are correct?

(A) f is many one but not even function
(B) Range of f contains two prime numbers
(C) f is aperiodic
(D) Graph of f does not lie below x–axis

माना $f : R \to R$ पर परिभाषित फलन $f(x) = \cos^{-1}(-\{-x\})$ है तब निम्न में से सत्य कथन होंगे : (जहां $\{x\}$ भिन्नात्मक मान फलन को दर्शाता है।)

(A) f बहुएकैकी लेकिन सम नहीं
(B) f के परिसर में दो अभाज्य संख्याएं हैं
(C) f आवर्ती फलन नहीं है
(D) f का आरेख x–अक्ष के नीचे नहीं होगा

Ans. ABD

Sol. We have $f(x) = \cos^{-1}(-\{-x\})$

D_f = all real

As $0 \le \{-x\} < 1 \ \forall x \in R$

$\Rightarrow \quad -1 < -\{-x\} \le 0$

So $\quad R_f = \left[\frac{\pi}{2}, \pi\right)$

Clearly, f is neither even nor odd.

But $\quad f(x+1) = f(x) \quad \Rightarrow \quad$ f is periodic with period 1.

11. The value of $\sum_{p=1}^{20}\sum_{q=1}^{20} 2\tan^{-1}\left(\frac{q}{p}\right)$ is :

$\sum_{p=1}^{20}\sum_{q=1}^{20} 2\tan^{-1}\left(\frac{q}{p}\right)$ का मान होगा :

(A) 200π
(B) 220π
(C) $\sum_{p=1}^{20}\sum_{q=1}^{20}\left(\tan^{-1}\frac{p}{q} + \tan^{-1}\frac{q}{p}\right)$
(D) 400π

Ans. AC

Sol. $\sum_{p=1}^{20}\sum_{q=1}^{20} 2\tan^{-1}\left(\frac{q}{p}\right)$

$$2\left[\tan^{-1}\left(\frac{1}{1}\right) + \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) + \ldots\ldots + \tan^{-1}\left(\frac{1}{20}\right) + \tan^{-1}\left(\frac{2}{1}\right)\right.$$

$$+\tan^{-1}\left(\frac{2}{2}\right)+\tan^{-1}\left(\frac{2}{3}\right)+.........+\tan^{-1}\left(\frac{2}{20}\right)+............$$

$$+\tan^{-1}\left(\frac{20}{1}\right)+\tan^{-1}\left(\frac{20}{2}\right)+\tan^{-1}\left(\frac{20}{3}\right)+.......+\tan^{-1}\left(\frac{20}{20}\right)\Bigg]$$

$$=2\times\tan^{-1}(1)\times 20+2\times\frac{\pi}{2}[19+18+.......+1]$$

$$=2\times\left(\frac{\pi}{4}\right)\times 20+2\times\frac{\pi}{2}\left[\frac{19\times 20}{2}\right]$$

$\Rightarrow 10\pi + 190\pi$

$= 200\pi$

(C) $$\sum_{p=1}^{20}\sum_{q=1}^{20}\left(\tan^{-1}\left(\frac{p}{q}\right)+\tan^{-1}\frac{q}{p}\right)$$

$$\sum_{p=1}^{20}\sum_{q=1}^{20}\left(\tan^{-1}\frac{p}{q}+\cot^{-1}\frac{p}{q}\right)$$

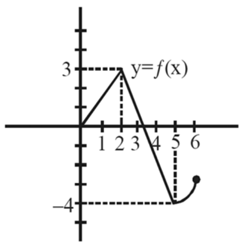
$$=\sum_{p=1}^{20}\sum_{q=1}^{20}\left(\frac{\pi}{2}\right)$$

$$=\sum_{p=1}^{20}\left(\frac{\pi}{2}+\frac{\pi}{2}+.......+\frac{\pi}{2}\right)=\sum_{p=1}^{20}(10\pi)$$

$\Rightarrow 10\pi+10\pi+.......+10\pi$

$= 200\pi$

12. The graph shows a part of an even function $f: [-6,6] \to R$, then :



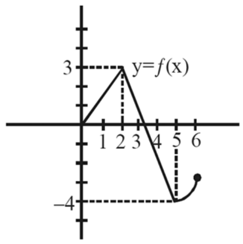
(A) Domain of $f(|x|+3)$ is $[-3,3]$

(B) Range of $f(|x|+3)$ is $\left[-4,\frac{2}{3}\right]$

(C) $\cos^{-1}\left(f\left(\frac{1}{3}\right)\right)+\cos^{-1}\left(f\left(\frac{43}{14}\right)\right)=\frac{2\pi}{3}$

(D) Number of solutions of the equation $f(f(x))=2$ is 4

दिया गया आरेख समफलन f: $[-6,6] \rightarrow R$ के एक भाग को दर्शाया हो, तो :



(A) $f(|x|+3)$ का प्रांत $[-3,3]$ है

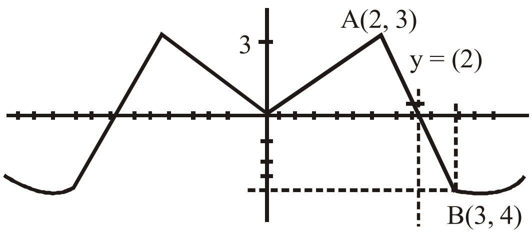
(B) $f(|x|+3)$ का परिसर $\left[-4,\frac{2}{3}\right]$ है

(C) $\cos^{-1}\left(f\left(\frac{1}{3}\right)\right)+\cos^{-1}\left(f\left(\frac{43}{14}\right)\right)=\frac{2\pi}{3}$

(D) समीकरण $f(f(x))=2$ के हलों की संख्या 4 है

Ans. ABC

Sol.



$f(x) \Rightarrow \text{domain } x \in [6, 6]$

$f(x) \Rightarrow x \in [0, 6]$

$f(|x|+3) \Rightarrow x \in [-3, 3]$

$AB \Rightarrow y-3=-\frac{7}{3}(x-2)$

$7x + 3y = 23$

$x = 3$

$y = 3 - \frac{7}{3}(3-2) = 3 - \frac{7}{3}$

$y_{max} = \frac{2}{3}$

$y_{min} = -4$

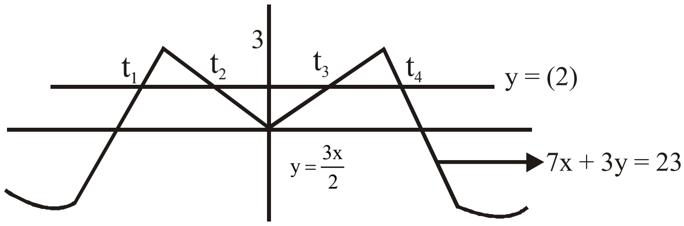
Range of $f(|x|+3) \Rightarrow \left[-4,\frac{2}{3}\right]$

(C) $\Rightarrow y=\frac{3}{2}x$

$$f\left(\frac{1}{3}\right)=\frac{1}{2} \qquad f\left(\frac{43}{14}\right)\Rightarrow y=3-\frac{7}{3}\left(\frac{43}{14}-2\right)=3-\frac{7}{3}\left(\frac{15}{14}\right)=\frac{1}{2}$$

$$\cos^{-1}\left(\frac{1}{2}\right)+\cos^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$$

(D) $f(f(x))=2$



$f(x)=t$

$f(t)=2$

$t_3 = 1.3 \qquad t_4 = 2.4$

$t_2 = -1.3 \qquad t_1 = -2.4$

$f(x)=t_1$, $f(x)=t_2$, $f(x)=t_3$, $f(x)=t_4$

$f(x)=-2.4 \qquad f(x)=-1.3 \qquad f(x)=1.3 \qquad f(x)=2.4$

Total no. of sol. = 12

13. Which of the following statement is/are true :

(A) $\cos^{-1}(\cos(1+|\sin x|)) = 1+|\sin x|$ for all real x

(B) $\sin^{-1}(\sin((\cos^2 x-\sin^2 x))) = \cos^2 x-\sin^2 x\ \forall\ x \in R$.

(C) $\tan^{-1}(\tan(x^2+2x+2)) = x^2+2x+2\ \forall\ x \in R$

(D) $\tan^{-1}(\tan(3\sin x)) = 3\sin x\ \forall\ x \in R$

निम्न में से कौनसा/कौनसे कथन सत्य है/होंगे :

(A) सभी वास्तविक x के लिए $\cos^{-1}(\cos(1+|\sin x|)) = 1+|\sin x|$

(B) $\sin^{-1}(\sin((\cos^2 x-\sin^2 x))) = \cos^2 x-\sin^2 x\ \forall\ x \in R$

(C) $\tan^{-1}(\tan(x^2+2x+2)) = x^2+2x+2\ \forall\ x \in R$

(D) $\tan^{-1}(\tan(3\sin x)) = 3\sin x\ \forall\ x \in R$

Ans. AB

Sol. (A) $\cos^{-1}(\cos(1+|\sin x|)) = 1+|\sin x|$

$\cos^{-1}(\cos[1,2])$

$1+|\sin x| \in [1,2]$

(B) $\sin^{-1}(\sin((\cos^2 x-\sin^2 x))) = \cos^2 x-\sin^2 x$

$$\sin^{-1}\left[\sin\left(\cos 2x\right)\right]$$

$$\Rightarrow \cos 2x \Rightarrow \cos^2 x-\sin^2 x$$

(C) $\tan^{-1}(\tan(x^2+2x+2)) = x^2+2x+2$

$\tan^{-1}\left(\tan\left((x+1)^2+1\right)\right) = x^2+2x+2$ for only $x \in \left[1, \frac{\pi}{2}\right]$

(D) $\tan^{-1}(\tan(3\sin x)) = 3\sin x$

$\tan^{-1}(\tan([-3, 3]))$

$\Rightarrow 3\sin x$; for only $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$